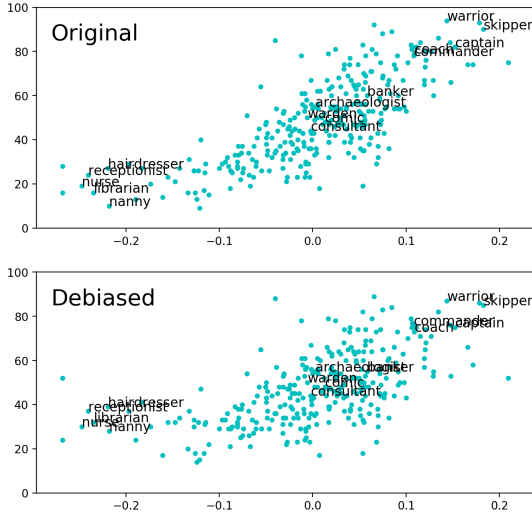
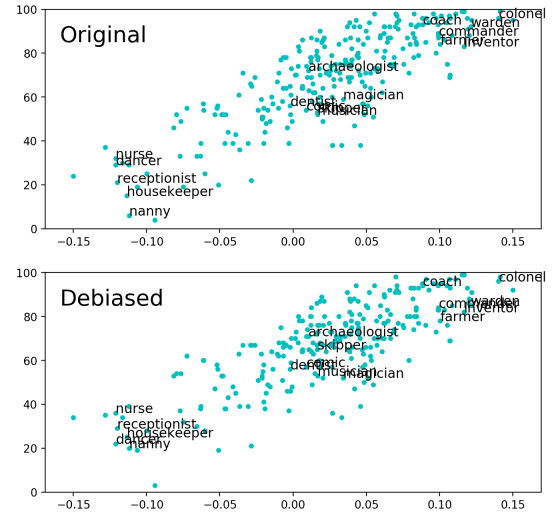




(a) The plots for HARD-DEBIASED embedding, before (top) and after (bottom) debiasing.



(b) The plots for GN-GLOVE embedding, before (top) and after (bottom) debiasing.

Figure 2: The number of male neighbors for each profession as a function of its original bias, before and after debiasing. We show only a limited number of professions on the plot to make it readable.

measure with the original bias measure. For the HARD-DEBIASED embedding we get a Pearson correlation of 0.686 (compared to a correlation of 0.741 when checking neighbors according to the biased version). For the GN-GLOVE embedding we get a Pearson correlation of 0.736 (compared to 0.773). All these correlations are statistically significant with p-values of 0.

**Professions** We consider the list of professions used in Bolukbasi et al. (2016b) and Zhao et al. (2018)<sup>11</sup> in light of the neighbours-based bias definition. Figure 2 plots the professions, with axis X being the original bias and axis Y being the number of male neighbors, before and after debiasing. For both methods, there is a clear correlation between the two variables.

We observe a Pearson correlation of 0.606 (compared to a correlation of 0.747 when checking neighbors according to the biased version) for HARD-DEBIASED and 0.792 (compared to 0.820) for GN-GLOVE. All these correlations are significant with p-values  $< 1 \times 10^{-30}$ .

**Association between female/male and female/male-stereotyped words** We replicate the three gender-related association experiments from Caliskan et al. (2017). For these experiments we use the full vocabulary since some of the words are not included in the reduced one.

The first experiment evaluates the association between female/male names and family and career words. The second one evaluates the association between female/male concepts and arts and mathematics words. Since the inherently gendered words (e.g. girl, her, brother) in the second experiment are handled well by the debiasing models we opt to use female and male names instead. The third one evaluates the association between female/male concepts and arts and science words. Again, we use female and male names instead.<sup>12</sup>

For the HARD-DEBIASED embedding, we get a p-value of 0 for the first experiment, 0.00016 for the second one, and 0.0467 for the third. For the GN-GLOVE embedding, we get p-values of  $7.7 \times 10^{-5}$ , 0.00031 and 0.0064 for the first, second and third experiments, respectively.

**Classifying previously female- and male-biased words** Can a classifier learn to generalize from some gendered words to others based only on their

<sup>11</sup><https://github.com/tolga-b/debiaswe/tree/master/data/professions.json>

<sup>12</sup>All word lists are taken from Caliskan et al. (2017): **First experiment:** *Female names:* Amy, Joan, Lisa, Sarah, Diana, Kate, Ann, Donna. *Male names:* John, Paul, Mike, Kevin, Steve, Greg, Jeff, Bill. *Family words:* home, parents, children, family, cousins, marriage, wedding, relatives. *Career words:* executive, management, professional, corporation, salary, office, business, career. **Second experiment:** *Arts Words:* poetry, art, dance, literature, novel, symphony, drama, sculpture. *Math words:* math, algebra, geometry, calculus, equations, computation, numbers, addition. **Third experiment:** *Arts words:* poetry, art, Shakespeare, dance, literature, novel, symphony, drama. *Science words:* science, technology, physics, chemistry, Einstein, NASA, experiment, astronomy.

representations? We consider the 5,000 most biased words according to the original bias (2,500 from each gender), train an RBF-kernel SVM classifier on a random sample of 1,000 of them (500 from each gender) to predict the gender, and evaluate its generalization on the remaining 4,000. For the HARD-DEBIASED embedding, we get an accuracy of 88.88%, compared to an accuracy of 98.25% with the non-debiased version. For the GN-GLOVE embedding, we get an accuracy of 96.53%, compared to an accuracy of 98.65% with the non-debiased version.

## 5 Discussion and Conclusion

The experiments described in the previous section reveal a systematic bias found in the embeddings, which is independent of the gender direction. We observe that semantically related words still maintain gender bias both in their similarities, and in their representation. Concretely, we find that:

1. Words with strong previous gender bias (with the same direction) are easy to cluster together.
2. Words that receive implicit gender from social stereotypes (e.g. receptionist, hairdresser, captain) still tend to group with other implicit-gender words of the same gender, similar as for non-debiased word embeddings.
3. The implicit gender of words with prevalent previous bias is easy to predict based on their vectors alone.

The implications are alarming: while suggested debiasing methods work well at removing the gender direction, the debiasing is mostly superficial. The bias stemming from world stereotypes and learned from the corpus is ingrained much more deeply in the embeddings space.

We note that the real concern from biased representations is not the association of a concept with words such as “he”, “she”, “boy”, “girl” nor being able to perform gender-stereotypical word analogies. While these are nice “party tricks”, algorithmic discrimination is more likely to happen by associating one implicitly gendered term with other implicitly gendered terms, or picking up on gender-specific regularities in the corpus by learning to condition on gender-biased words, and generalizing to other gender-biased words (i.e., a resume classifier that will learn to favor male over

female candidates based on stereotypical cues in an existing—and biased—resume dataset, despite of being “oblivious” to gender). Our experiments show that such classifiers would have ample opportunities to pick up on such cues also after debiasing w.r.t the gender-direction.

The crux of the issue is that the gender-direction provides a way to *measure* the gender-association of a word, but *does not determine* it. Debiasing methods which directly target the gender-direction are for the most part merely hiding the gender bias and not removing it. The popular definitions used for quantifying and removing bias are insufficient, and other aspects of the bias should be taken into consideration as well.

## Acknowledgments

This work is supported by the Israeli Science Foundation (grant number 1555/15), and by the Israeli ministry of Science, Technology and Space through the Israeli-French Maimonide Cooperation program.

## References

- Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, and Adam Kalai. 2016a. Man is to computer programmer as woman is to homemaker? debiasing word embeddings. *arXiv:1607.06520*.
- Tolga Bolukbasi, Kai-Wei Chang, James Y Zou, Venkatesh Saligrama, and Adam T Kalai. 2016b. Man is to computer programmer as woman is to homemaker? debiasing word embeddings. In *Advances in Neural Information Processing Systems*.
- Aylin Caliskan, Joanna J Bryson, and Arvind Narayanan. 2017. Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334):183–186.
- Nikhil Garg, Londa Schiebinger, Dan Jurafsky, and James Zou. 2018. Word embeddings quantify 100 years of gender and ethnic stereotypes. *Proceedings of the National Academy of Sciences*, 115(16):E3635–E3644.
- Laurens van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-sne. *Journal of machine learning research*, 9(Nov):2579–2605.
- Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*.

Jeffrey Pennington, Richard Socher, and Christopher D Manning. 2014. Glove: Global vectors for word representation. In *Proceedings of EMNLP*.

Brian Hu Zhang, Blake Lemoine, and Margaret Mitchell. 2018. Mitigating unwanted biases with adversarial learning. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society*.

Jieyu Zhao, Yichao Zhou, Zeyu Li, Wei Wang, and Kai-Wei Chang. 2018. Learning gender-neutral word embeddings. In *Proceedings of EMNLP*, pages 4847–4853.